

Ziyi Guo

ziyiguo59@gmail.com | 781-652-1324
zg908.github.io

RESEARCH INTERESTS

Computational mechanisms of learning and decision-making; neural population dynamics and neural-behavioral relationships. Prospective postdoctoral interests include human neurotechnology, computational psychiatry, and clinically relevant neural biomarkers.

EDUCATION

Johns Hopkins University School of Medicine
Ph.D. in Neuroscience | Baltimore, MD

Expected May 2027

Brandeis University
B.S. in Computer Science, Neuroscience, and Psychology | Waltham, MA
GPA: 4.0

May 2021

RESEARCH EXPERIENCE

Hussain Shuler Laboratory, Johns Hopkins
Graduate Research Assistant | Reinforcement learning and decision-making

Jun 2022 - Present

- Investigate time-evolving signals in the anterior cingulate cortex using a reinforcement learning framework.
- Develop reinforcement learning model variants to predict decision-making behavior and neural decoders in TensorFlow to model neural activity.
- Develop Python workflows for processing and statistical analysis of more than 50 TB of Neuropixels recording data.

Jadhav Laboratory, Brandeis University
Undergraduate Research Assistant | Learning and memory

Jan 2018 - May 2021

- Developed MATLAB workflows to process and integrate neural recordings from the prefrontal cortex and hippocampus.
- Trained DeepLabCut neural network models for automated behavioral feature extraction and analysis.

PEER-REVIEWED PUBLICATION

Young, R. A., Shin, J. D., **Guo, Z.**, & Jadhav, S. P. (2025). Hippocampal-prefrontal communication subspaces align with behavioral and network patterns in a spatial memory task. *eNeuro*, 12(9), ENEURO.0336-24.2025.
doi:10.1523/ENEURO.0336-24.2025

TRANSLATIONAL RESEARCH EXPERIENCE

Red Abbeyes Labs, Lutherville, MD
NeuroBehavior Analyst Intern | Biotech and pharmaceutical development

Feb 2025 - May 2025

- Developed behavioral testing protocols for postoperative dementia treatments.
- Automated video analysis for a contract research organization (CRO), improving efficiency by 50%.
- Contributed to business planning with multiple CRO partners.

HONORS AND RESEARCH SUPPORT

Class of 1955 Prize in Creative Ability	2021
Phi Beta Kappa, elected as a junior	2020
Computational Neuroscience Traineeship, NIDA training (\$15,000)	2020
M. R. Bauer Foundation Summer Undergraduate Research Fellowship (\$5,000)	2019

CONFERENCE PRESENTATIONS AND ACCEPTED ABSTRACTS

Guo, Z., & Hussain Shuler, M. G. (2026). Time-evolving value signals in the ACC and M2 during foraging. Oral presentation (student speaker), Johns Hopkins Neuroscience Retreat, Baltimore, MD.

Guo, Z., Cassidy, R., & Hussain Shuler, M. G. (2025). Time-evolving signals in the anterior cingulate cortex under a reinforcement learning framework. Johns Hopkins Neuroscience Retreat, Janelia, VA. Presentation.

Guo, Z., Young, R., & Jadhav, S. P. (2021, January). Communication subspaces for local-field potential defined network states. SfN Global Connectome: A Virtual Event. Presentation.

Guo, Z., Chen, Y.-L., & Youngstrom, E. A. (2020). Intergenerational transmission of parenting style: discontinuities predicted by culture. Accepted abstract, American Psychological Association Annual Convention.

Guo, Z., Young, R., & Jadhav, S. P. (2019). Got Milk? Assessing goal-directed behavior driven by sensory cues and memory in rats. Brandeis University SciFest, Waltham, MA. Presentation.

Guo, Z., Chen, Y.-L., & Youngstrom, E. A. (2019). Intergenerational transmission of parenting style in a cross-cultural college student sample. American Psychological Association Annual Convention, Chicago, IL. Presentation.

TEACHING AND MENTORSHIP

Research mentorship, Johns Hopkins

Jun 2022 - Present

- Mentor undergraduate and rotation students in research.

Graduate Teaching Assistant, Johns Hopkins

Spring 2023

- Supported Neuroscience and Cognition II by coordinating lecture materials with faculty, improving slide accessibility, and holding office hours.

Teaching Assistant, Computer Science, Brandeis

Jun 2019 - May 2020

- Led a team of 16 teaching assistants supporting Java programming courses with more than 100 students; designed object-oriented programming assignments and automated grading tests.

TECHNICAL EXPERTISE

Programming: Python, R, SQL, MATLAB, Java.

Neural and behavioral analysis: Neuropixels data processing, time-series analysis, neural decoding, statistical analysis, data visualization, large-scale data processing.

Modeling and machine learning: Reinforcement learning models of behavior, neural networks, TensorFlow, PyTorch, DeepLabCut.

Tools: Git, SSH, Unity.

LEADERSHIP AND SERVICE

Team Leader, Virtual Reality Anthropology Library

Mar 2019 - May 2021

Brandeis Skunkworks

- Coordinated an interdisciplinary team and anthropology collaborators; developed interactive artifact displays using Unity and Oculus, and supported artifact scanning and processing.

Treasurer, Neuroscience Club, Brandeis

Jan 2018 - May 2019

- Managed funding, organized faculty panels for general audiences, and participated in student-led journal clubs.